

nano-archimedes: A GNU Package for Nano-World Simulations

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In March issue of EFY Plus we discussed Archimedes, a Monte Carlo simulator for various semiconductor device simulations. In this issue we will talk about nano-archimedes, yet another technology computer aided design (TCAD) software. The tool is based on Wigner equation, a convenient formulation of quantum mechanics in terms of phase-space and two many-body approaches, density functional theory (DFT) and a novel time-dependent *ab-initio* method.

Generally speaking, nano-archimedes can be used for the simulation of various technology-relevant situations that involve dynamics of electrons, such as transport in nanometer-scale semiconductor devices (for example, nanodevices) as well as dynamics of many-body problems in quantum chemistry (for example, molecular electronics).

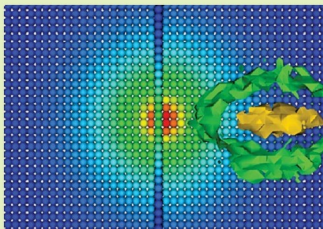
The source code, written in C language, is a truly cross-platform code that can be compiled on a huge variety of machines (serial and parallel) without any particular effort. The first release of the software was made early this year under GPL version 3.

Jean Michel Sellier, the main developer of nano-archimedes, put up every effort to free up more codes on quantum simulations so that a new user could have advanced researches in this area, instead of spending time coding from scratch.

Where nano-archimedes finds its applications

Before going deep into the approaches used for this software, let us take a look at some areas where nano-archimedes could be conveniently used.

Simulating post-CMOS designs. We are now in the so-called post-



Simulation of a wave packet in proximity of Coulombic potential using nano-archimedes

complementary metal oxide semiconductor (CMOS) era. With the continuous process of miniaturisation, the semiconductor's active length has reduced to nanometers. At these scales quantum effects get prominent and the behaviour is quite different from what classical devices (like CMOS transistors) were designed for. Thus we require a new design paradigm that exploits the typical phenomena of quantum mechanics. The gap between theoretical comprehension and rapidly-advancing experiments can be effectively bridged with the help of quantum simulations using this software.

Practical design and optimisation of realistic solotronic devices.

With advances in technology, modern electronics have the capability of manipulating single dopant atoms in semiconductor materials with atomic precision. These advancements took the name and shape as a unique branch of electronics called solotronics. Again, the theoretical comprehension cannot keep in pace with the experimental advancements that would eventually prevent the practical design and optimisation

of solotronic devices. There is an increased demand for TCAD software that would help in a meaningful study of solotronic designs, which require a time-dependent, full quantum, multi-dimensional model, even in the relatively simpler case of ballistic regime. nano-archimedes is a free

open source option that could ideally, and easily, fill up this space.

Simulating chemical systems. Researchers working in applied atomic physics and quantum chemistry try simulating quantum chemical systems like atoms and molecules with the help of quantum mechanical models. For determination of chemical properties of such a system, their electronic structure should be calculated. This involves numerical simulation of the quantum many-body problem, one of the most computationally-demanding and difficult problems in applied physics.

A pseudo-potential model computes the effective potential consisting of a super position of the core electrons and nucleus potentials. This model can thus be used to reduce the original problem to the simulation of valence electrons only. This modelling could be effectively done by Wigner formalism used in this software.

Ab-initio simulations of the quantum many-body problem

The term *ab-initio* simulations refers to simulations based on the first